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Given data:

$[NH_3] = 1.8 \times 10^{-2}\,M$

$[N_2] = 1.2 \times 10^{-2}\,M$

$[H_2] = 3 \times 10^{-2}\,M$

$K_c = ?$

Solution:

$$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$$

$$K_C = \frac{[NH_3]^2}{[N_2][H_2]^3} = \frac{1.8\times10^{-2}\times1.8\times10^{-2}}{1.2\times10^{-2}\times3\times10^{-2}\times3\times10^{-2}\times3\times10^{-2}}$$

$$= 1\times10^{3}\ l^2\ mol^{-2}$$

2. The equilibrium constant at 298 K for a reaction is 100.

$A + B \rightleftharpoons C + D$

If the initial concentration of all the four species is 1 M, the equilibrium concentration of D (in mol lit^{-1}) will be

Given data:

$[A] = [B] = [C] = [D] = 1\ M$

$K_c = 100$

$[D]_{eq} = ?$

Solution:

Let x be the no moles of reactants reacted

	A	B	C	D
Initial concentration	1	1	1	1
At equilibrium (as per reaction stoichiometry	1-x	1-x	1+x	1+x

$$K_C = \frac{[C][D]}{[A][B]}$$

$$100 = \frac{(1+x)(1+x)}{(1-x)(1-x)}$$

$$\sqrt{100} = \sqrt{\frac{(1+x)(1+x)}{(1-x)(1-x)}}$$

$$10 = \frac{1+x}{1-x}$$

$$10(1-x) = 1+x$$

$$10-10x-1-x = 0$$

$$9-11x = 0$$

$$11x = 9$$

$$x = \frac{9}{11} = 0.818$$

$[D]\ eq = 1+x = 1 + 0.818 = 1.818M$

Evaluate Yourself

4) 1 mol of PCl_5, kept in a closed container of volume 1 dm^3 and was allowed to attain equilibrium at 423 K. Calculate the equilibrium composition of reaction mixture. (The K_c value for PCl_5 dissociation at 423 K is 2)

8.8 Le-Chatelier's Principle

Many chemical reactions that have industrial importance such as synthesis of ammonia are reversible in nature. It is important to know the reaction conditions to produce maximum yield. If a system at equilibrium is disturbed by modifying the reaction conditions, then the system adjusts itself the new conditions to re-establish the equilibrium.

The effect of change in reaction conditions such as temperature, pressure and concentration etc… on a system at equilibrium can be predicted by Le Chatelier-Braun principle.